Real Time Exit Blockage Detection via Door-Centric Change Monitoring for Emergency Evacuation

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Abstract: The availability of unobstructed exits is crucial for ensuring safe and efficient evacuation during emergencies. However, real-time exit blockage detection, especially in buildings with multiple exits, poses a critical challenge, and dynamic unforeseen obstructions can cause confusion, delays, and even injuries as evacuees may be directed toward inaccessible routes. Human monitoring is often proven unreliable under such rapidly changing conditions. In addition, existing vision-based approaches rely on exhaustive obstacle catalogs and struggle to adapt to new and unforeseen blockage scenarios. In this work, we propose a novel, door-centric change detection model that bypasses the need for object-specific obstacle recognition. Instead of identifying all possible obstructions, our method focuses solely on monitoring the visual state of doors. Using a lightweight YOLOv11s model, we track the bounding box geometry of each door and detect blockages based on deviations from a known unobstructed reference. This approach simplifies the detection process while improving adaptability to unknown obstruction types. We evaluate our model on diverse and previously unseen video streams containing various types of door blockage. The results demonstrate high accuracy in estimating accessible door areas and show strong generalization capabilities beyond the limitations of traditional object detection methods. Notably, the model supports real-time inference on low-power edge devices, such as the Raspberry Pi, enabling practical deployment in resource-constrained environments.

SCIENCE AND TECHNOLOGY PUBLICATIONS

1 INTRODUCTION

Recent disasters and dynamic hazards in multi-floor, multi-exit buildings pose significant challenges for evacuation management. The architectural complexity and extensive network of egress points in such structures can overwhelm individual spatial awareness, making it difficult to identify the least congested and most accessible routes without comprehensive, real-time information on exit availability (**?**). Traditional evacuation procedures rely on trained personnel manually guiding occupants to safety; however, during large-scale emergencies, such staff may be scarce or overwhelmed. Moreover, rapidly evolving conditions—such as earthquake debris, spreading fires, or toxic gas leaks—can quickly render pre-planned escape routes impassable, endangering both evacuees and first responders (**?**).

Sensor-based monitoring systems have offered partial mitigation by continuously collecting environmental data and reporting anomalies to a central command unit (**?**). However, many such "black-box" sensors lack the spatial context necessary to localize or classify unforeseen hazards—such as a collapsed beam obstructing a corridor or smoke rapidly accumulating near an exit—without explicit reports from personnel on the ground (**?**). Recent advances in machine learning (ML), and particularly in deep convolutional neural networks, have enabled robust computer vision solutions capable of scene understanding in indoor and urban environments (**?**). These systems can detect and localize critical objects or conditions in real time, delivering actionable insights directly to evacuation management platforms (**?**). Yet, existing vision-based approaches typically rely on exhaustive catalogs of obstacle classes and struggle to adapt when novel or unexpected blockages arise (**?**).

To address these limitations, we propose a novel

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door-centric change-detection model that monitors only the visual state of exits. By training a lightweight YOLOv11 network to track each door's bounding-box geometry and comparing it against a stored unobstructed reference, our model signals potential blockages whenever significant deviations occur. This strategy eliminates the need for object-specific obstacle recognition, simplifying the detection pipeline while enhancing adaptability to unknown obstruction types.

Our approach treats the doorway as the primary object of analysis instead of classifying individual obstacles. The model infers exit blockage through variations in the door's bounding-box geometry and IoU consistency over time. By combining distributed supervision with IoU-weighted objectives, the approach achieves high sensitivity to partial occlusions while maintaining real-time performance on low-power edge devices (e.g., Raspberry Pi). We evaluate the method on diverse, previously unseen video streams featuring multiple door-blockage scenarios and observe strong generalization beyond traditional object detectors, with accurate estimation of accessible door areas and practical deployability in resource-constrained environments.

The remainder of this paper is organized as follows. Section **??** reviews prior work in evacuation support and computer vision for obstruction detection. Section **??** details our door-state change model, data preparation, and training procedures. Section **??** presents quantitative and qualitative evaluations under diverse conditions. Finally, Section **??** concludes with a summary of findings and potential directions for future research..

2 RELATED WORK

Automated evacuation monitoring has drawn on both sensor-based and vision-based technologies. Sensor arrays (e.g., environmental detectors for smoke, gas, or structural strain) offer continuous anomaly reporting but lack the spatial resolution required to pinpoint obstructions without human intervention (**?**). In contrast, recent advances in deep convolutional networks have enabled real-time scene understanding in complex indoor and urban contexts, facilitating object localization under challenging lighting and occlusion conditions (**?**; **?**). Lightweight detectors, such as YOLOv11 variants, have been shown to achieve a favorable balance of speed and accuracy on resource-constrained hardware (**?**).

General object detection research has addressed limited-label regimes through semi- and weakly supervised approaches. Semi-supervised methods leverage large pools of unlabeled images to boost accuracy with few annotations (**?**), while weakly supervised frameworks integrate contextual cues via interactive learning to compensate for coarse image-level labels (**?**). Although these techniques improve robustness in noisy or sparsely annotated domains, they still depend on explicit class definitions and struggle when novel obstacle types appear.

Open-world change detection extends beyond fixed-category recognition by focusing on state transitions. (**?**) investigate video-based object state changes without exhaustive object catalogs, demonstrating improved adaptability to unforeseen scenarios. This perspective aligns with our door-state monitoring paradigm, which infers obstructions through deviations in bounding-box geometry rather than through direct obstacle classification.

Despite progress in multi-modal fusion (e.g., radar–camera (**?**), LiDAR–image (**?**)) and compressed-domain detection (**?**), few studies target the specific problem of exit blockage in evacuation contexts. To our knowledge, no prior work has proposed a door-centric change-detection model that eschews object-specific recognition in favor of monitoring bounding-box alterations.

Recent studies have shown that robust object state monitoring relies not only on the detection backbone but also on the optimization strategy that governs how classification confidence and localization quality are coupled. Generalized Focal Loss (GFL) unifies these two aspects by integrating class confidence and IoU quality into a single objective, replacing discrete bounding-box targets with continuous probability distributions optimized via Quality and Distribution Focal Loss (**?**). This approach of learning distributed rather than point targets has proven effective across various tracking and detection domains. For instance, Gaussian Distributed Focal Loss has stabilized localization quality estimation in STARK-style trackers by introducing a Gaussian-shaped supervision over continuous bounding-box quality scores (**?**).

Building on this foundation, several studies have proposed adaptive loss formulations. Automated Focal Loss dynamically adjusts the focusing parameter γ according to the batchwise prediction quality, reducing manual tuning and accelerating convergence (**?**), while dynamic loss mechanisms further emphasize complex samples to improve convergence in one-stage detectors (**?**). On the regression side, IoU-based loss families have evolved to address limitations of overlap-dependent gradients. Focal-EIoU combines efficient IoU decomposition with focal weighting to prioritize high-quality anchors (**?**), and Focal IoU loss

introduces adaptive weighting to better learn difficult bounding boxes (**?**). Geometric refinements such as Control Distance IoU (CDIoU) (**?**) and Central-Gaussian Angle-IoU (CA-IoU) (**?**) improve convergence speed and alignment precision by incorporating spatial distance and angular penalties, while hybrid approaches like GCIoU integrate complementary terms to enhance both accuracy and stability during training (**?**).

Together, these developments highlight two core principles underlying this research: (i) decoupling recognition from state monitoring through geometry- and quality-oriented supervision (**?**; **?**), and (ii) leveraging IoU-aware regression losses that remain informative under occlusion, partial overlap, or irregular obstruction shapes (**?**; **?**; **?**; **?**).

3 METHOD

In this work, we propose an innovative solution that evaluates door-state changes to detect potential blockages. Traditional object detection methods localize and classify each object in an image or video stream by returning its bounding-box coordinates. When two bounding boxes overlap, this may indicate collision or occlusion between the corresponding objects. However, to identify every conceivable obstruction, especially in emergency scenarios where any physical object might block an exit, the model would need to be trained on an impractically large variety of items.

To overcome this limitation, we focus exclusively on a single object class, namely the "door," which serves as a proxy for exit viability. Rather than detecting all possible occluding objects, our approach monitors changes in the door's bounding box coordinates and dimensions. By comparing the inferred bounding box of the door in a current frame against a stored reference (healthy) state, the system can infer the presence of an obstruction whenever significant deviations occur.

3.1 Dataset Collection

Our method requires a diverse set of door images, encompassing both exterior and interior contexts. A review of existing benchmarks, such as COCO, revealed the absence of a standardized door dataset. Although Google Open Images V7 (**?**) contains door annotations, the labels are not domain-specific: they include automobile doors, telephone-kiosk doors, gates, and more. From the 5,916 images labeled as "door" in Open Image (**?**), we manually selected 457 unique photographs depicting building entrance doors. To enrich interior variability, we supplemented this with 181 additional images of indoor doors sourced via internet search. In total, our curated dataset comprises 648 unique door images.

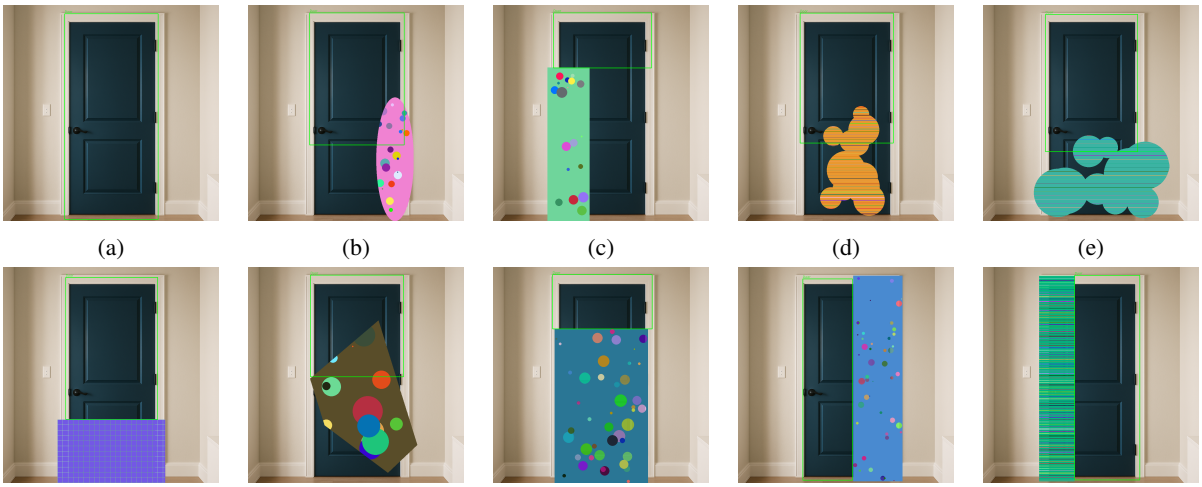
3.2 Synthetic Blockage Generation

To simulate a wide range of obstructions, we generated nine blocked variants for each original (healthy) door image, yielding a corpus of 6,480 frames. Using the PIL (Pillow) library's `Image` and `ImageDraw` modules, we programmatically overlaid shapes of varying sizes, patterns, and colors to mimic plausible blockage scenarios. Figure **??** provides an overview of these manipulations: **??** shows an unobstructed door; **??** and **??** illustrate blockages in the bottom right and bottom left quadrants, respectively; **??** depicts a central obstruction at the door base; Figures **??** through **??** simulate horizontal occlusions affecting both side frames at varying heights; and **??** and **??** represent vertical blockages extending across full side frames with partial coverage of the top and bottom edges.

3.3 Annotation Protocol

Each synthesized image was annotated following a consistent protocol. For unobstructed doors, the bounding box encompasses the entire door frame. When blockage occurs on opposite sides, either horizontally (left and right) or vertically (top and bottom), the annotation is restricted to the visible portion of the door up to the occluded frame. In cases of unilateral obstruction, the bounding box is either expanded beyond the visible frame (when the occluding shape is small, as in Figure **??**) or confined to the unobstructed segment of the door (when the occluding shape is large, as in Figure **??**). Annotated bounding boxes are depicted in green in Figure **??**.

During the annotation process, we deliberately ignored any visual elements situated centrally on the door surface, such as decorative panels, graffiti, or wreaths. We operated under the principle that genuine obstructions must intersect the door's structural boundaries, specifically, the left or right stiles, the top rail, or the door sill, thereby indicating contact with the frame or the floor. By excluding centrally located patterns from our annotations, we prevent the model from misattributing non-obstructive decorations as blockages. Moreover, this annotation strategy emphasizes the primacy of the door frame regions in the learning objective, guiding the model to assign greater significance to alterations along the frame edges rather than to variations in the door's central



(a) (b) (c) (d) (e)

(f) (g) (h) (i) (j)

Figure 1: Illustrative door conditions: (a) unobstructed baseline; (b)–(d) obstructed on the right, left, and center; (e)–(h) varying degrees of horizontal obstruction; (i)–(j) vertical obstruction.

panel.

3.4 Corpus Augmentation

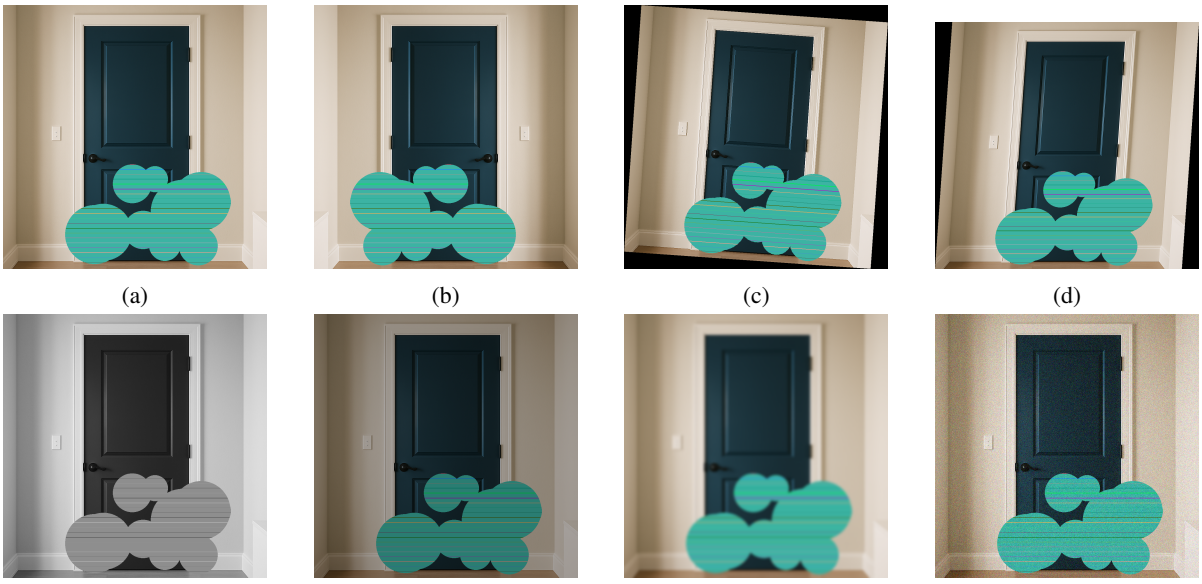
To enhance model robustness against variability in lighting, perspective, and image quality, we applied a suite of augmentation techniques to our synthesized door dataset. Although the original door images were collected under diverse illumination regimes, ranging from natural daylight to nocturnal artificial lighting, and captured from multiple viewing angles, real-world deployments may encounter additional noise factors. Consequently, we introduced systematic transformations designed to emulate common distortions and environmental conditions. Figure **??** provides a visual summary of these augmentations. In particular, Figure **??** shows the baseline synthesized door without modification; Figure **??** demonstrates a horizontal mirror flip to simulate mirrored camera installations; Figures **??** and **??** depict rotations of $\pm 4^{\circ}$ and affine shearing, respectively, which model slight camera misalignments; Figure **??** converts the image to grayscale to test performance under monochromatic capture; and Figure **??** applies a randomized brightness reduction of up to 15%, reflecting overcast or poorly lit conditions. Finally, Figures **??** and **??** illustrate Gaussian blurring and additive pixel-level noise, respectively, to simulate motion blur and sensor artifacts encountered in low-visibility or high-speed scenarios.

These augmentation procedures were applied at random to each of the 6,480 synthesized images, comprising both the original "healthy" door frames and their nine blocked variants, to produce two expanded corpora of differing scale. The first augmented set contains 16,871 images, while the second comprises 21,915 images. By training and evaluating our model on these two datasets, we aim to assess the impact of augmentation volume on detection accuracy and generalization. Specifically, we investigate whether a larger, more heterogeneous training corpus yields statistically significant improvements in the model's ability to detect door-state anomalies under previously unseen conditions.

3.5 Model Training

To detect the "door" class in both static images and live video frames, we adopted the YOLOv11s architecture, a one-stage detector optimized for deployment on resource-constrained edge devices. YOLO networks are renowned for their real-time performance, and recent studies have demonstrated their suitability in challenging environments. For example, (**?**) evaluated YOLO's throughput and accuracy on embedded hardware, concluding that its single-stage pipeline effectively balances speed and precision under tight computational budgets. Similarly, (**?**) applied YOLO to underwater imagery, an analog to emergency conditions characterized by low contrast and cluttered backgrounds, and reported robust detection of small, partially occluded objects. More recently, (**?**) benchmarked the YOLO11s variant on mobile platforms, highlighting its adaptability and low-latency inference even when processing complex scenes. Based on these findings, we selected YOLOv11s for its combination of compact model size, rapid inference, and proven resilience in noisy visual contexts.

We conducted four distinct training models to



(a) (b) (c) (d)

(e) (f) (g) (h)

Figure 2: Variations of a door image used for robustness analysis: (a) original door image, (b) flipped horizontally, (c) rotation distortion, (d) skewed image, (e) grayscale transformation, (f) brightness change, (g) blurriness, and (h) noise corruption.

evaluate how corpus size and stopping criteria affect model sensitivity to door-state changes. The summary configuration of these four models' training is presented in Table **??**. In the first model (A), we trained YOLOv11s on the larger augmented corpus (21,915 images) for up to 100 epochs, employing early stopping with a patience parameter of ten epochs. Early stopping halts training if validation precision and loss fail to improve over ten successive epochs, thereby mitigating overfitting and enhancing generalization (**?**; **?**). The best validation weights were obtained at epoch 31, after which training ceased. While this model achieved high confidence in detecting unobstructed doors, its performance degraded on instances with partial frame occlusions, indicating insufficient representation of subtle state changes.

To address this shortfall, we retrained on the same larger corpus for 200 epochs without early stopping (model B), saving model checkpoints at 50-epoch intervals. This procedure yielded five candidate models (at epochs 50, 100, 150, 200), plus an intermediate best performer at epoch 73, selected for its optimal trade-off between localization loss and detection accuracy. Notably, the checkpointed models at epochs 50, 73, and 100 demonstrated markedly improved recall of doors with unilateral frame blockages, as well as increased confidence scores on these challenging examples.

Next, we investigated the impact of corpus scale by training on the smaller augmented set (16,871 images). We ran a 100-epoch session without early stopping (model C), recording the best weights at epoch 75, strikingly close to the optimal epoch from the larger corpus. This smaller-corpus model outperformed the second large-corpus model (model B) in terms of bounding-box sensitivity to partial obstructions, despite exhibiting a higher Distributed Focal Loss (DFL). As formulated by (**?**), DFL encourages the network to concentrate its probability mass around the continuous ground-truth target location, yielding tighter and more precise box predictions. (**?**) further demonstrated that elevated focal-loss weighting can enhance localization accuracy by driving the model to emphasize boundary-proximal regions.

Table 1: Summary of Model Training Configurations.

Model	Training Configuration
A	Trained on the larger augmented corpus for 100 epochs, with early stopping applied (patience of ten epochs).
B	Trained on the larger augmented corpus for 200 epochs without early stopping; checkpoints saved at epochs 50, 100, 150, and 200, with the best-performing checkpoint selected.
C	Trained on the smaller augmented corpus for 100 epochs without early stopping, using validation metrics to identify the optimal epoch.
D	Trained on a randomly augmented version of the smaller corpus (same size) for 100 epochs to confirm the reproducibility of performance trends observed in Model C.

Under traditional interpretation, an increasing or stable DFL across training epochs may signal overfitting, as the network "memorizes" training samples rather than generalizing (**?**). However, our objective diverges from pure object detection: we aim to maximize sensitivity to door-state deviations. The smaller corpus, with its reduced redundancy of augmented instances, appears to accentuate the model's responsiveness to bounding-box perturbations, even as DFL rises. To validate this hypothesis, we generated a new random augmentation of equivalent size to the smaller dataset and performed a fourth training run for 100 epochs (model D). The best weights again emerged at epoch 77, mirroring previous optimal checkpoints. Consistently, the model exhibited high DFL values accompanied by superior performance in detecting door-frame alterations.

Having established the training regimen and identified the configurations that best capture door-state changes, we now turn to Section **??** to present the quantitative and qualitative evaluation of our models under diverse test scenarios.

4 RESULTS

Model A, which employed early stopping, terminated training at epoch 31. Although this model reliably detected and localized fully unobstructed doors, it proved ineffective at identifying partially or fully blocked exits. Model D was constructed to verify the reproducibility of Model C's performance: we applied a randomized augmentation pipeline of identical size and an alternative train–validation split, yielding results consistent with those of Model C. Consequently, we focus our subsequent analysis on Model B (trained on the larger corpus as seen in Fig. **??**) and Model C (trained on the smaller corpus as seen in Fig **??**).

We first examine the training dynamics, loss behavior, and class-level performance of our models trained on the large (Model B) and small (Model C) augmented corpora (Figure **??** and Figure **??**). We then transition to an evaluation on entirely unseen video sequences to assess real-time blockage detection capabilities.

4.1 Large-Corpus Model Performance

Figures **??** summarize the training and validation curves for our YOLOv11s models trained on the large (21,915 images) corpus. It presents the evolution of box-loss, classification-loss, Distributed Focal Loss (DFL), and the primary detection metrics: precision, recall, *m*AP50, and *m*AP50–95. The figure shows both training and validation boxes, and classification losses decline smoothly, indicating effective learning of door localization and classification. The DFL decreases steadily from approximately 1.3 to 0.9, reflecting increasingly compact predicted bounding-box distributions around ground-truth targets. Precision and recall both converge near 0.995, while *m*AP50 exceeds 0.99 and *m*AP50–95 plateaus around 0.92 by epoch 60.

The confusion matrix in Figure **??** (left) further confirms these findings: out of 2,048 true "door" instances, the model correctly predicts 2,023 (99.8% true positives), with only 5 missed detections (false negatives). There are 25 false positives (background predicted as door) and no true background examples in the validation set, underscoring the model's strong discrimination when ample training data are available.

4.2 Small-Corpus Model Performance and DFL Behavior

Similarly, for the small corpus (Figure **??**), the training DFL again decreases monotonically, but the validation DFL follows a U-shaped curve: it falls from around 1.29 to a minimum of 1.10 by epoch 30, then rises back to 1.40 by epoch 100. Traditional interpretation would view this resurgence as overfitting, coinciding with a lower *m*AP50–95 (peaking near 0.78 versus 0.92 for the large corpus), despite a high *m*AP50 of 0.98.

The corresponding confusion matrix (Figure **??**, right) shows 1,425 true positives out of 1,439 "door" instances (99.0% recall), with 14 false negatives and 73 false positives. The elevated false-positive rate reflects the smaller training set's reduced exposure to background variability.

Crucially, the increase in validation DFL for the small-corpus model aligns with improved sensitivity to partial obstructions. As DFL rises, the network concentrates its probability mass more narrowly around the visible edges of the ground-truth box. This tighter localization enables the model to generate bounding boxes that exclude occluded regions, thereby accurately delineating the unobstructed portion of the door, exactly the behavior required for reliable blockage alerts. Thus, while the small-corpus model exhibits traditional signs of overfitting in its DFL curve, it paradoxically better fulfills our primary objective of detecting door-state changes.

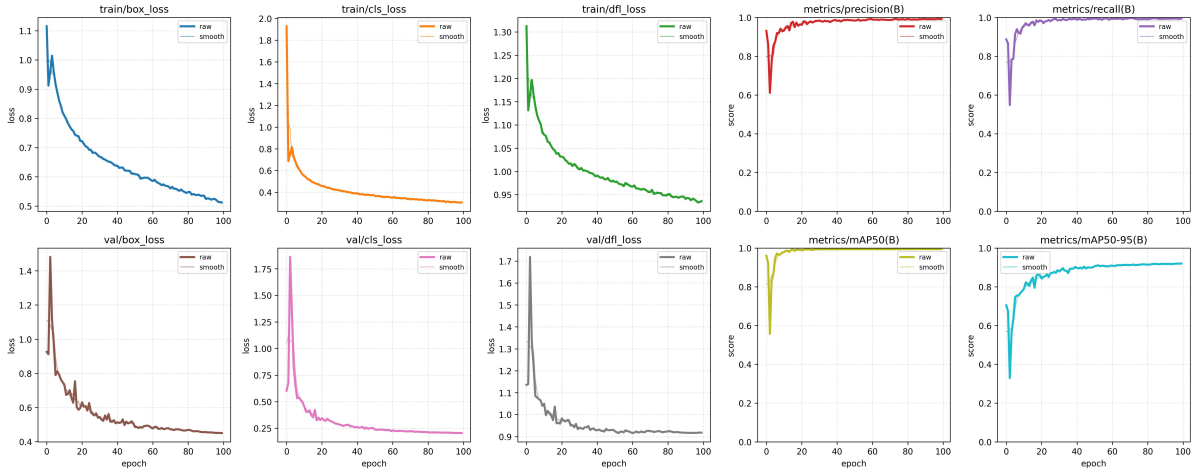


Figure 3: Model Training & Validation Results on Large Corpus.

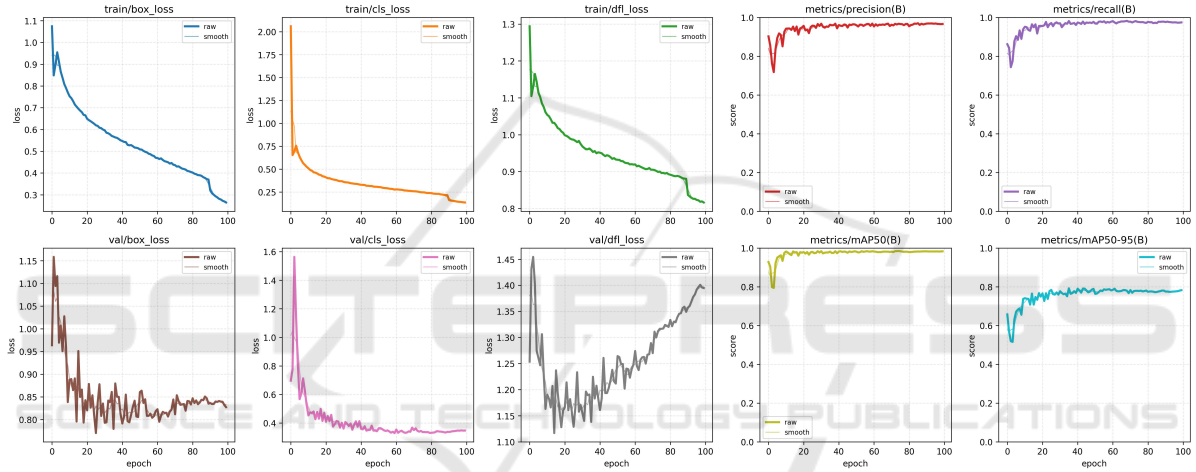


Figure 4: Model Training & Validation Results on Small Corpus.

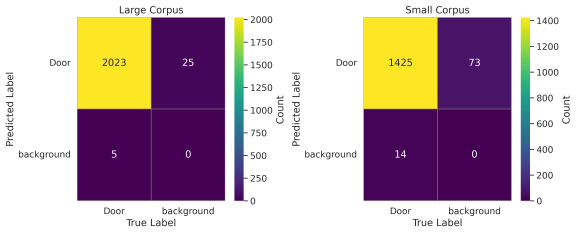


Figure 5: Confusion Matrix for Large and Small Corpora Models.

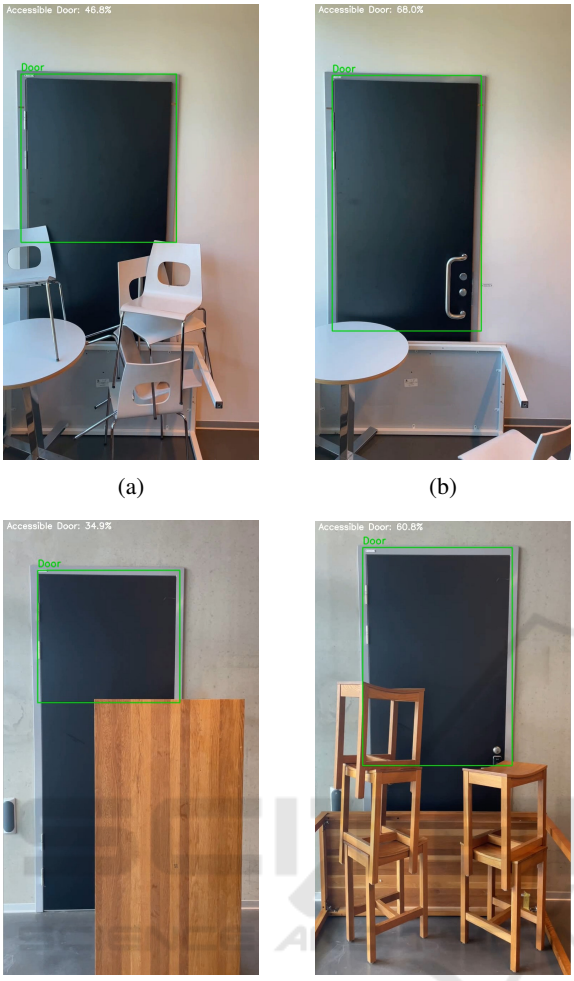
4.3 Model Evaluation on Unseen Data

To assess real-time blockage detection under realistic conditions, we recorded two videos on the Malmö University campus in which doors were successively obstructed by chairs, tables, and benches. None of these frames appeared in our training or validation sets. We performed two complementary tests: a "static" evaluation on a manually annotated image subset, and a "dynamic" evaluation directly on the video streams.

4.3.1 Static Test on Raspberry Pi 5

From the two videos, we extracted and hand-annotated approximately 300 frames to create a ground-truth test dataset. We deployed the YOLOv11-small model trained on the small corpus, chosen for its superior change-sensitivity, onto a Raspberry Pi 5. Since the default PyTorch runtime is not optimized for ARM architectures, we converted the model into four additional formats:

- **ONNX:** an open interchange format that enables interoperability between frameworks;
- **OpenVINO:** Intel's toolkit for optimizing and accelerating inference on CPU and integrated GPUs;
- **MNN:** Alibaba's lightweight inference engine designed for mobile and edge devices;



(a) (b)

(c) (d)

Figure 6: Example outputs of the model on two unseen video streams.

- **NCNN:** Tencent's high-performance neural network inference framework for ARM processors.

Table **??** compares each format in terms of model size, detection accuracy measured by *m*AP50–95, average inference time per image, and resulting frames per second (FPS). All formats maintain comparable localization performance (0.8365–0.8433 *m*AP50–95), indicating that format conversion does not degrade accuracy. However, inference speed varies markedly: the PyTorch model processes at only 0.75 FPS (1326 ms/frame), whereas ONNX achieves 2.24 FPS, MNN 3.72 FPS, NCNN 4.84 FPS, and OpenVINO leads with 5.78 FPS (172.9 ms/frame). These results illustrate that OpenVINO and NCNN offer the best trade-off between throughput and precision for on-edge door-state monitoring.

Table 2: Model Format Performance Comparison.

Format	Size (MB)	metrics/mAP50-95(B)	Inference time (ms/im)	FPS
PyTorch (default)	18.3	0.8416	1326.29	0.75
ONNX	36.2	0.8419	447.12	2.24
OpenVINO	36.3	0.8365	172.93	5.78
MNN	36.1	0.8428	268.64	3.72
NCNN	36.1	0.8433	206.80	4.84

4.3.2 Dynamic Video-Stream Test

For live evaluation, we first computed a reference bounding box by running the model on an unobstructed frame. As each obstruction was introduced, we measured the ratio of the predicted visible-door area to this reference area, yielding a per-frame "accessible door" percentage. Figure **??** presents four representative frames:

(a) Stacked chairs reduce the accessible area to 46.8%;

(b) A partially blocking table yields 68.0% visible door;

(c) A table obstruction leaves only 34.9% exposed;

(d) Multiple benches produce a 60.8% clearance.

In all cases, the green bounding box accurately conforms to the unobstructed region, demonstrating the model's ability to generalize to novel obstruction types and to quantify exit viability in real time.

Together, these evaluations confirm that our change-detection model not only generalizes to previously unseen environments but also operates with sufficient speed and accuracy on edge hardware to support live evacuation monitoring.

5 CONCLUSION AND FUTURE WORK

In this paper, we have presented a novel computer-vision framework for detecting exit blockages by monitoring door–state changes rather than attempting exhaustive object cataloging. Leveraging a curated dataset of 648 unique door images and nine synthetically generated obstruction variants per image, we trained a lightweight YOLOv11s model on both large and small augmented corpora. Our analysis of training dynamics—particularly the behavior of the Distributed Focal Loss—revealed a trade-off between generalization and sensitivity to partial occlusions: models trained on smaller datasets exhibited tighter bounding-box localization around visible door regions, while larger-corpus models achieved superior overall *m*AP50–95 scores.

We further demonstrated real-time applicability on a Raspberry Pi 5 by benchmarking multiple inference formats (PyTorch, ONNX, OpenVINO, MNN,

NCNN) and achieving up to 5.78 FPS with minimal loss in detection accuracy. Finally, evaluations on two previously unseen video streams confirmed that our system accurately quantifies the visible door area under a variety of novel obstruction scenarios, yielding reliable "accessible door" measurements.

While these results substantiate the viability of change-detection for evacuation monitoring, future work should explore larger pools of unique door images—spanning diverse architectural styles, materials, and lighting conditions—and more varied obstruction scenarios (e.g., debris of irregular shapes, dynamic crowding). Such expanded testing will be essential to validate the robustness and generalizability of our approach across the full spectrum of real-world emergency environments.

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